StimMAS: A Multi-Agent Framework for Automated Linguistic Stimulus Construction in Psychological Research

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Online demo: https://huggingface.co/spaces/CUHKLPL/stimuli_generator

Introduction. The construction of high-quality experimental stimuli is foundational to psychological and psycholinguistic research, yet it remains labor-intensive, expertise-dependent, and difficult to scale. As the field increasingly moves toward large-scale, cross-linguistic, and rapidly iterated experimental designs, stimulus construction has become a major methodological bottleneck. Recent advances in large language models (LLMs) offer new possibilities for automating stimulus generation, but existing implementations rely largely on single-agent pipelines, raising concerns about quality control, internal validity, and reproducibility. We introduce **StimMAS**, the first **multi-agent LLM framework** for automated linguistic stimulus construction, designed to formalize and modularize generation, validation. We test whether stimuli produced by StimMAS (i) replicate established psycholinguistic effects and (ii) improve stimulus quality relative to single-agent generation.

Methods. StimMAS consists of three specialized LLM agents: a **Generator**, which produces candidate stimuli given an experimental design and exemplars; a **Validator**, which enforces experiment-specific linguistic and semantic constraints and rejects invalid or redundant items; and a **Scorer**, which assigns quantitative ratings along relevant psycholinguistic dimensions (e.g., predictability, plausibility). Human researchers supervise the pipeline at critical decision points. We validated StimMAS by generating novel stimulus sets for three established psycholinguistic paradigms and testing them against human-authored materials in behavioral experiments with 300 native English speakers: (1) Sound–shape association (nonword judgments; N = 60), (2) Word length and predictivity (sentence completion; N = 120), and (3) Implausible sentence interpretation (rational inference under noise; N = 120). Across experiments, stimulus source (human-authored vs. LLM-constructed) was manipulated between items and participants. Trial-level responses were analyzed using logit mixed-effects models. An additional **ablation study** compared the full multi-agent pipeline to a single-agent system, and validator decisions were evaluated against expert judgments.

Results. Across all three experiments, **LLM-constructed stimuli reliably replicated the target effects** reported in prior work. Sound–shape associations, predictivity-driven preference for shorter words, and structure-sensitive nonliteral interpretations were all observed with StimMAS-generated materials (all $ps < .001$). Crucially, no effect was eliminated by replacing human-authored stimuli with LLM-generated ones. Item-level analyses further showed that LLM-generated stimuli elicited **lower response entropy** than human-authored items, indicating more consistent interpretations across participants, while maintaining comparable item-level effect-size distributions. The ablation study demonstrated that **removing the Validator agent** substantially degraded stimulus quality, increasing repetition and semantic violations. Validator judgments showed moderate-to-strong agreement with expert evaluations (F1 = .69 for GPT-4o), and markedly outperformed single-agent pipelines.

Conclusion. These findings demonstrate that **multi-agent LLM systems**, when embedded in a human-supervised workflow, can generate linguistic stimuli that are experimentally valid, theoretically aligned, and reproducible. By decomposing stimulus construction into generation, validation, and scoring, StimMAS addresses key limitations of single-agent approaches and provides a scalable and transparent tool for integrating AI into the empirical workflows of psychological science, lowering barriers to high-quality experimental design without replacing expert judgment.

Reference

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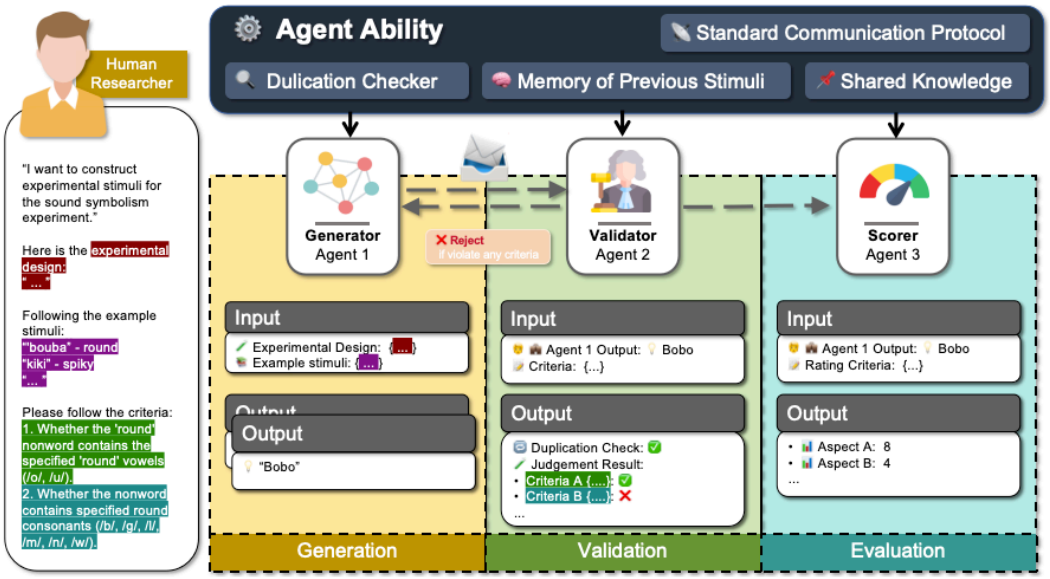

Figure 1. Multi-agent system for automated stimuli generation and evaluation in psychological experiments. The system consists of three specialized agents: a Generator (Agent 1), a Validator (Agent 2), and a Scorer (Agent 3). The Generator creates candidate stimuli (e.g., *Bobo* for a sound symbolism experiment) based on the experimental design and exemplars. The Validator enforces quality by checking for duplication and ensuring each stimulus meets experiment specific criteria (e.g., semantic validity); non-compliant outputs are rejected and sent back for regeneration. Approved stimuli proceed to the Scorer, which assigns quantitative ratings along different psycholinguistic dimensions (e.g., predictability, contextual fit).

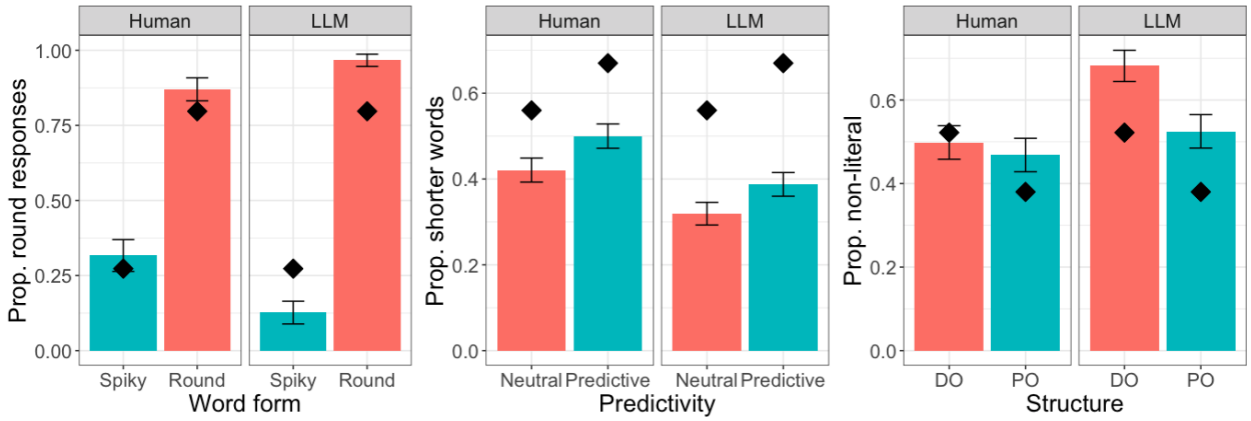

Figure 2. Results across three experiments comparing original and LLM-constructed stimuli.